

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 13: Statistics

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mean of grouped data

1. The data below has a mean of 35 and a total frequency of 25. Find x and y . **CBSE 2026, 5 marks**

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	1	x	5	7	y	3	1

SOLUTION

Total frequency: $17 + x + y = 25$, so $x + y = 8 \dots (1)$

Use the class marks 5, 15, 25, 35, 45, 55, 65.

$$\sum f_i x_i = 5 + 15x + 125 + 245 + 45y + 165 + 65 = 605 + 15x + 45y$$

$$\text{Mean} = \frac{605 + 15x + 45y}{25} = 35, \text{ so } 15x + 45y = 270, \text{ that is } x + 3y = 18 \dots (2)$$

Subtracting (1) from (2): $2y = 10$, so $y = 5$ and $x = 3$.

Answer: $x = 3, y = 5$

2. The mean of the data below is 18. Find the missing frequency f , and then find the mode.

CBSE 2025, 5 marks

Class	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Frequency	7	6	9	13	f	5	4

SOLUTION

Use the class marks 12, 14, 16, 18, 20, 22, 24.

$$\sum f_i = 44 + f$$

$$\sum f_i x_i = 84 + 84 + 144 + 234 + 20f + 110 + 96 = 752 + 20f$$

$$\frac{752 + 20f}{44 + f} = 18, \text{ so } 752 + 20f = 792 + 18f \text{ and } f = 20.$$

The highest frequency is now 20, so the modal class is 19–21 with $l = 19, h = 2, f_1 = 20, f_0 = 13, f_2 = 5$.

$$\text{Mode} = 19 + \frac{20 - 13}{40 - 13 - 5} \times 2 = 19 + \frac{14}{22}$$

Answer: $f = 20$; **mode** ≈ 19.64

3. Observations $x_1, x_2, \dots, x_n$ have frequencies $f_1, f_2, \dots, f_n$ and mean $\bar{x}$. What is the value of $\sum f_i(x_i - \bar{x})$? (a) $n\bar{x}$ (b) 1 (c) $\sum f_i$ (d) 0 **CBSE 2024, 1 mark**

SOLUTION

$$\sum f_i(x_i - \bar{x}) = \sum f_i x_i - \bar{x} \sum f_i$$

Since $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, we have $\bar{x} \sum f_i = \sum f_i x_i$, so the value is 0.

Answer: (d)

4. Every observation in a set of data is increased by 3. What happens to the mean? (a) it does not change (b) it increases by 3 (c) it increases by 6 (d) it increases by $3n$ **CBSE 2023, 1 mark**

SOLUTION

The total increases by $3n$, so the mean increases by $\frac{3n}{n} = 3$.

Answer: (b)

2 Empirical relationship

5. The mean of some data is 12 and its mode is 21. Find the median. (a) 6 (b) 13.5 (c) 15 (d) 14 **CBSE 2026, 1 mark**

SOLUTION

$$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean} = 21 + 24 = 45$$

$$\text{Median} = 15$$

Answer: (c)

6. For a set of data, the mode is $15x$ and the mean is $18x$. Find the median. **CBSE 2025, 1 mark**

SOLUTION

$$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean} = 15x + 36x = 51x$$

$$\text{Answer: Median} = 17x$$

3 Median of grouped data

7. When data is arranged in order, the observation in the middle is called the: (a) mode (b) median (c) mean (d) deviation **CBSE 2024, 1 mark**

SOLUTION

The median is the middle value of data arranged in order.

Answer: (b)

8. In a game of BINGO the host has balls numbered 1 to 75 and calls out the number on a ball picked at random. In one game 48 balls were called, as shown in the table. (i) Name the median class. (ii) Find the probability that the first ball called had an even number. (iii) Find the median of the data, OR find its mode. **CBSE 2024, 4 marks, case study**

Numbers announced	0–15	15–30	30–45	45–60	60–75
Number of times	8	9	10	12	9

SOLUTION

(i) $n = 48$, so $\frac{n}{2} = 24$. The cumulative frequencies are 8, 17, 27, 39, 48. The first one greater than 24 is 27, so the median class is 30–45.

(ii) Of the numbers 1 to 75, the even numbers are 2, 4, ..., 74, which is 37 numbers. Probability = $\frac{37}{75}$

(iii) Median = $l + \frac{\frac{n}{2} - cf}{f} \times h = 30 + \frac{24 - 17}{10} \times 15 = 30 + 10.5 = 40.5$

OR: the modal class is 45–60, with $f_1 = 12$, $f_0 = 10$, $f_2 = 9$. Mode = $45 + \frac{12 - 10}{24 - 10 - 9} \times 15 = 45 + 6 = 51$

Answer: (i) 30–45 (ii) $\frac{37}{75}$ (iii) median 40.5, or mode 51

9. The table shows the monsoon rainfall recorded in 24 sub-divisions of the country. (i) Write the modal class. (ii) Find the median, OR find the mean rainfall. (iii) A sub-division with at least 1000 mm of rain is said to have good rainfall. How many had good rainfall? **CBSE 2023, 4 marks**

Rainfall (mm)	200–400	400–600	600–800	800–1000	1000–1200	1200–1400	1400–1600	1600–1800
Sub-divisions	2	4	7	4	2	3	1	1

SOLUTION

(i) The highest frequency is 7. **Modal class: 600–800.**

(ii) Cumulative frequencies: 2, 6, 13, 17, 19, 22, 23, 24. $n = 24$, so $\frac{n}{2} = 12$ and the median class is 600–800, with $l = 600$, $cf = 6$, $f = 7$, $h = 200$.

Median = $600 + \frac{12 - 6}{7} \times 200 = 600 + 171.43$. **Median ≈ 771.43 mm.**

OR: with class marks 300, 500, ..., 1700, $\sum f_i x_i = 600 + 2000 + 4900 + 3600 + 2200 + 3900 + 1500 + 1700 = 20400$. Mean = $\frac{20400}{24}$. **Mean = 850 mm.**

(iii) Classes from 1000 mm upwards: 2 + 3 + 1 + 1. **7 sub-divisions had good rainfall.**

4 Mode of grouped data

10. The table gives the marks of 80 students in a test. Find the modal class. (a) 10–20 (b) 20–30 (c) 30–40 (d) 50–60 **CBSE 2023, 1 mark**

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of students	3	12	27	57	75	80

SOLUTION

The table is cumulative. Subtract to get the frequency of each class:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	3	9	15	30	18	5

The highest frequency is 30, in the class 30–40.

Answer: (c)

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